Cambridge (CIE) IGCSE Maths: Extended

Your notes

Probability Diagrams (Tree & Venn Diagrams)

Contents

Your notes

Two-Way Tables

Two Way Tables

What are two-way tables?

- **Two-way tables** are tables that compare **two types** of **characteristics**
  - For example, a college of 55 students has two year groups (Year 12 and Year 13) and two language options (Spanish and German)
  - The two-way table is shown:

	Spanish	German
Year 12	15	10
Year 13	5	25

How do I find probabilities from a two-way table?

- Draw in the **totals** of each **row** and **column**
  - Include an **overall total** in the bottom-right corner
    - It should be the sum of the totals above, or to its left (both work)
  - For the example above:

	Spanish	German	*Total*
Year 12	15	10	*25*
Year 13	5	25	*30*
Total	*20*	*35*	*55*

- Use this to answer **probability** questions
  - If a random student is selected from the whole college, it will be out of 55

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- The probability a student selected from the college studies Spanish and is in Year 12 is $\frac{15}{55}$
- The probability a student selected from the college studies Spanish is $\frac{20}{55}$
- If a random student is selected from a **specific category**, the **denominator** will be that **category total**
  - The probability a student selected from Year 13 studies Spanish is $\frac{5}{30}$

Examiner Tips and Tricks

- Check your row and column totals add up to the overall total, otherwise all your probabilities will be wrong!

Worked Example

At an art group, children are allowed to choose between colouring, painting, clay modelling and sketching.

A total of 60 children attend and are split into two classes: class A and class B.
12 of class A chose the activity colouring and 13 of class B chose clay modelling.
A total of 20 children chose painting and a total of 15 chose clay modelling.
8 of the 30 children in class A chose sketching, as did 4 children in class B.

(a) Construct a two-way table to show this information.

Read through each sentence and fill in the numbers that are given

	Colouring	Painting	Clay modelling	Sketching	Total
Class A	12			8	30
Class B			13	4	
Total		20	15		60

Use the row and column totals to fill in any obvious missing numbers

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	Colouring	Painting	Clay modelling	Sketching	Total
Class A	12		15 - 13 = 2	8	30
Class B			13	4	60 - 30 = 30
Total		20	15	8 + 4 = 12	60

Use the row and column totals again to find the last few numbers

	Colouring	Painting	Clay modelling	Sketching	Total
Class A	12	30 - 12 - 2 - 8 = 8	2	8	30
Class B	30 - 12 - 13 - 4 = 1	20 - 8 = 12	13	4	30
Total	12 + 1 = 13	20	15	12	60

Write out your final answer

	Colouring	**Painting**	**Clay modelling**	**Sketching**	**Total**
Class A	**12**	**8**	**2**	**8**	**30**
Class B	**1**	**12**	**13**	**4**	**30**
Total	**13**	**20**	**15**	**12**	**60**

(b) Find the probability that a randomly selected child

(i) chose colouring,

(ii) is in class A, who chose sketching.

(i) We are not interested in whether the child is in class A or B
A total of 13 children chose colouring, out of 60 children

P(colouring) = $\frac{13}{60}$

(ii) 8 children in class A chose sketching
There are 60 children to select from

P(class A and sketching) = $\frac{8}{60} = \frac{2}{15}$

(c) A child in class B is selected at random. Find the probability they chose painting.

As we are only selecting from class B, this will be out of 30 (rather than the total of 60)
12 in class B chose painting

$$\text{P(painting, from class B only)} = \frac{12}{30} = \frac{2}{5}$$

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Probabilities from Venn Diagrams

Your notes

Probability & Venn Diagrams

How do I find probabilities from Venn diagrams?

- Count the **number** of elements you want and **divide** by the **total number** of elements
- For the Venn diagram shown below,
  - The probability of being **in *A*** is $\frac{5}{11}$
    - There are 5 elements in A out of 11 in total
  - The probability of being in **both** A **and** B is $\frac{2}{11}$
    - There are 2 elements in A and B (the intersection)
  - The probability of being **in *A***, but **not *B***, is $\frac{3}{11}$
    - 3 elements are in A but not B
- Some harder questions are **not** out of the **total** number, but out of a **restricted** number
  - The probability of being in B, **given that** you are already in A, is $\frac{2}{5}$
    - You are only interested in elements in A
    - There are 5 elements in A, out of which only 2 are also in B

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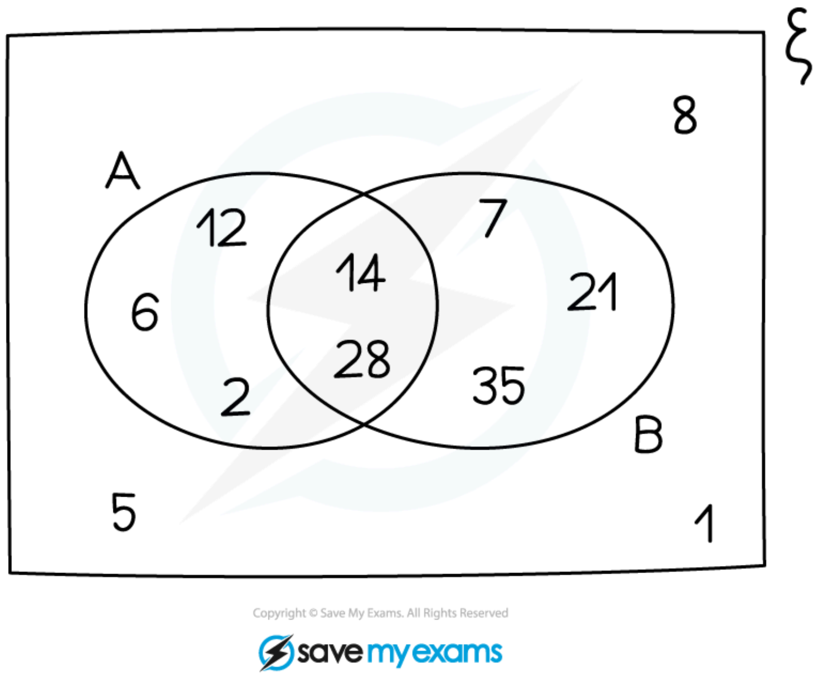

Examiner Tips and Tricks

- Be careful when filling in numbers for a Venn diagram
  - Some of the given numbers may need to be split between two sections of the Venn diagram
- Suppose 10 people have a cat, 8 people have a dog and 6 people have both a cat and a dog
  - Out of the 10 people who have a cat
    - 6 also have a dog
    - 4 do not have a dog
  - Out of the 8 people who have a dog
    - 6 also have a cat
    - 2 do not have a cat

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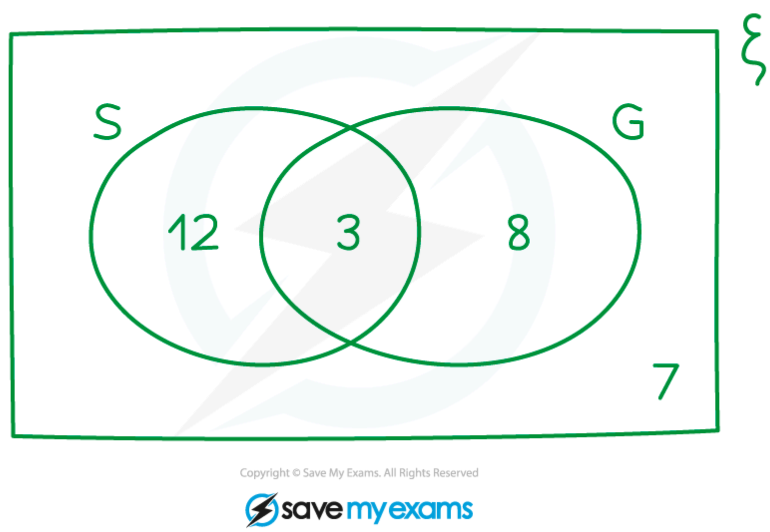
Worked Example

In a class of 30 students, 15 students study Spanish and 3 of the Spanish students also study German.
7 students study neither Spanish nor German.

(a) Draw a Venn diagram to show this information.

Draw the Venn diagram with its rectangular box and two (labelled) overlapping circles
3 students study both Spanish and German, so start here and work outwards
12 must study Spanish but not German (to get 15 in total for Spanish)
7 study neither, so this goes outside of the circles
To get 30 in total, 8 must study German but not Spanish



(b) Use your Venn diagram to find the probability that a student, selected at random from the class, studies Spanish but not German.

It helps to highlight Spanish but not German

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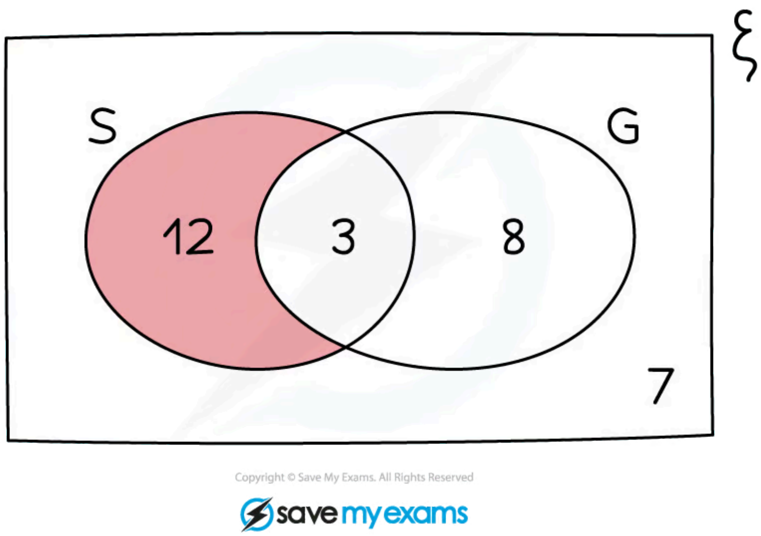

Divide the number of students studying Spanish but not German by the total number of students

Students studying Spanish but not German = 12
Total number of students = 30

$$\textbf{P(Spanish but not German)} = \frac{12}{30}\left(=\frac{2}{5}\right)$$

Probability Tree Diagrams

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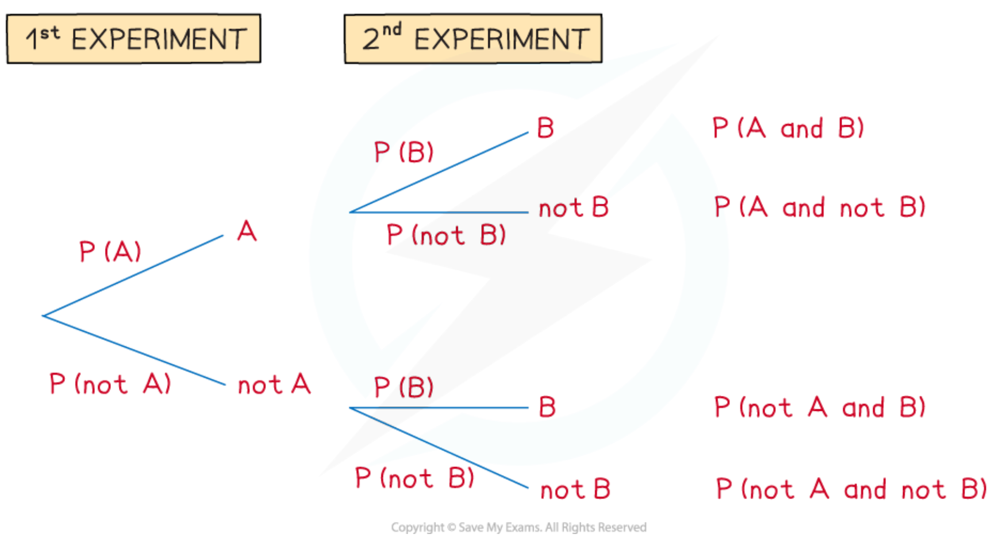
Tree Diagrams

How do I draw a tree diagram?

- **Tree diagrams** can be used for **repeated experiments** with **two outcomes**
  - The **1st experiment** has outcome **A** or **not A**
  - The **2nd experiment** has outcome **B** or not **B**
- Read the tree diagram from **left to right** along its **branches**
  - For example, the top branches give A followed by B
    - This is called A **and** B



How do I find probabilities from tree diagrams?

- Write the **probabilities** on each branch
  - Remember that P(**not** A) = 1 - P(A)
    - Probabilities on each **pair** of branches **add to 1**
- **Multiply** along the branches from **left to right**

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- This gives P(1st outcome **and** 2nd outcome)
- **Add** between the **separate cases**
  - For example
    - P(AA **or** BB) = P(AA) + P(BB)
- The probabilities of **all** possible cases **add to 1**
- If asked to find the probability of **at least one** outcome, it is quicker to do **1 - P(none)**

How do I use tree diagrams with conditional probability?

- Probabilities that depend on a particular thing having happened first in a tree diagram are called **conditional probabilities**
- For example, the probability that a team wins a game may **depend** on whether they won or lost the previous game
  - The **probabilities** for 'win' on the first set of branches may be **different** to those for 'win' on the second set of branches
- Another example of conditional probabilities is **"without replacement"** scenarios
  - e.g. two items are drawn from a bag of different coloured items without the first item drawn being replaced
  - The **probabilities** on the second set of branches will **change** depending on which branch has been followed on the first set of branches
    - The **denominators** in the probabilities for the second set of branches will be **one less** than those on the first set of branches
    - The **numerators** on the second set of branches will also change
- Conditional probability questions are sometimes introduced by the expression **'given that...'**
  - e.g. 'Find the probability that the team win their next game given that they lost their previous game'
- The **notation** $\mathrm{P}(A|B)$ is often used for conditional probabilities
  - That is read as 'the probability of *A* given *B*'
  - e.g. $\mathrm{P}(\text{win}|\text{lose})$ is the probability a team wins, given that they lost the previous game

Examiner Tips and Tricks

- When multiplying along branches with fractions, **don't cancel fractions** in your working

- Having the same denominator makes them easier to add together!

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Worked Example

A worker drives through two sets of traffic lights on their way to work.
Each set of traffic lights has only two options: green or red.

The probability of the first set of traffic lights being on green is $\frac{5}{7}$.

The probability of the second set of traffic lights being on green is $\frac{8}{9}$.

(a) Draw and label a tree diagram. Show the probabilities of every possible outcome.

Work out the probabilities of each set of traffic lights being on red, R
Use P(red) = 1 - P(green)

$$\text{P}(1^{\text{st}}\ \text{R}) = 1 - \text{P}(1^{\text{st}}\ \text{G}) = 1 - \frac{5}{7} = \frac{2}{7}$$
$$\text{P}(2^{\text{nd}}\ \text{R}) = 1 - \text{P}(2^{\text{nd}}\ \text{G}) = 1 - \frac{8}{9} = \frac{1}{9}$$

Draw the branches (with a label of G or R on the ends)
Write the probabilities above each branch
Calculate probabilities of each outcome by multiplying along the branches from left to right

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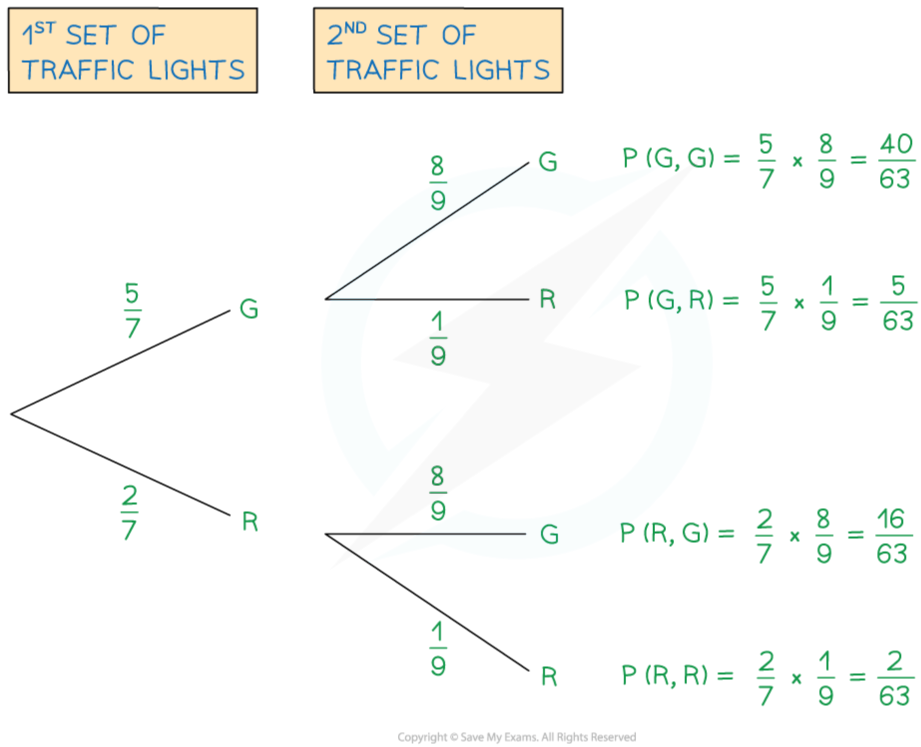

(b) Find the probability that both sets of traffic lights are on red.

This is the answer for P(R, R) from the tree diagram

$$\frac{2}{63}$$

(c) Find the probability that at least one set of traffic lights are on red.

<u>Method 1</u>

This means the 1st is green and the 2nd is red
Or the 1st is red and the 2nd is green
Or the 1st is red and the 2nd is red ('at least one' could mean both)

Add between the separate cases

$$\text{P(at least one R)} = \text{P(G, R)} + \text{P(R, G)} + \text{P(R, R)} = \frac{5}{63} + \frac{16}{63} + \frac{2}{63} = \frac{23}{63}$$

$$\frac{23}{63}$$

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Method 2

At least one red means all the possible cases shown except two greens

So P(at least 1 red) = 1 - P(two greens)

$$1 - P(G, G) = 1 - \frac{40}{63} = \frac{23}{63}$$

$$\frac{23}{63}$$

Worked Example

Liana has 10 pets. She has 7 guinea pigs (G) and 3 rabbits (R).

Liana is choosing two pets to feature in her latest online video.

First she is going to choose one of the pets at random. Once she has carried that pet to her video studio she is going to go back and choose a second pet at random to also feature in the video.

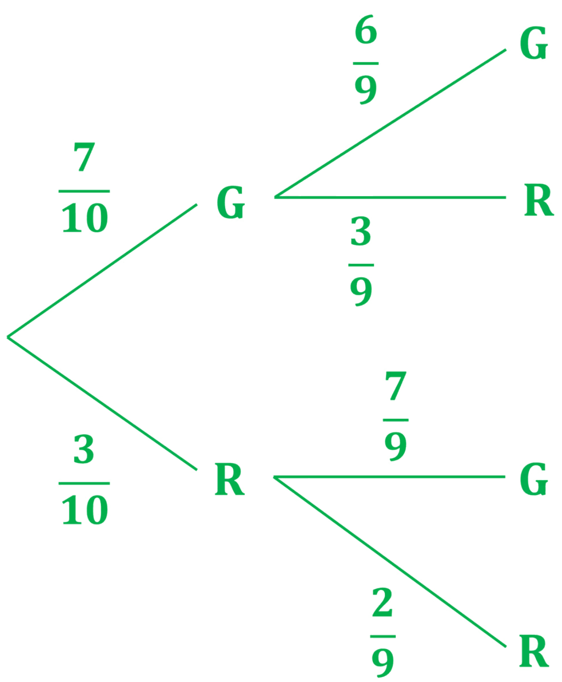
(a) Draw and label a tree diagram including the probabilities of all possible outcomes.

For the 1st pet chosen, there will be a 7/10 probability of choosing a guinea pig, and a 3/10 probability of choosing a rabbit.

If the first pet is a guinea pig, there will only be 6 guinea pigs and 3 rabbits left (9 animals total)
So for the second pet the probability of choosing a guinea pig would be 6/9, and probability of choosing a rabbit would be 3/9

If the first pet is a rabbit, there will only be 7 guinea pigs and 2 rabbits left (9 animals total)
So for the second pet the probability of choosing a guinea pig would be 7/9, and probability of choosing a rabbit would be 2/9

Put these probabilities into the correct places on the tree diagram, and then multiply along the branches to find the probabilities for each outcome



$$P(G, G) = \frac{7}{10} \times \frac{6}{9} = \frac{42}{90}$$

$$P(G, R) = \frac{7}{10} \times \frac{3}{9} = \frac{21}{90}$$

$$P(R, G) = \frac{3}{10} \times \frac{7}{9} = \frac{21}{90}$$

$$P(R, R) = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90}$$

(b) Find the probability that Liana chooses two rabbits.

As we have already calculated this probability in the table, we can just write the answer down

$$\mathbf{P(two\ \ rabbits) = P(R,\ R) = \frac{6}{90}\left(= \frac{1}{15}\right)}$$

(c) Find the probability that Liana chooses two different kinds of animal.

This would be "G AND R" OR "R AND G" so we need to add two of the final probabilities

$$\text{P(two different kinds)} = P(G, R) + P(R, G) = \frac{21}{90} + \frac{21}{90} = \frac{42}{90}$$

$$\mathbf{P(two\ \ different\ \ kinds) = \frac{42}{90}\left(= \frac{7}{15}\right)}$$

Combined Probabilities

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Combined Probability

How do I calculate combined probabilities?

- You can calculate probabilities of **one event after another** without needing tree diagrams
  - These are called **combined** (or successive) probabilities
- There are two **rules** to learn
  - **And** means **multiply** and **or** means **add**
  - P(A **and** B) = P(A) x P(B)
  - P(AA **or** BB) = P(AA) + P(BB)
- Try to **rephrase** each question using and / or
  - For example, when flipping a coin twice:
    - P(two heads) = P(head **and** head)
    - P(both the same) = P(head **and** head **or** tail **and** tail) = P(HH) + P(TT)
- Remember that P(**not** A) = 1 - P(A)

Worked Example

A box contains 3 blue counters and 8 red counters.
A counter is taken at random and its colour is noted.
The counter is put back into the box.
A second counter is then taken at random, and its colour is noted.

Work out the probability that

(a) both counters are red,

P(both red) = P(red and red)
This is P(red) × P(red) using the 'and rule'

$$\frac{8}{11} \times \frac{8}{11}$$

$$\frac{64}{121}$$

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(b) the two counters are of different colours.

P(different colours) = P(blue and red or red and blue)
This is P(B and R) + P(R and B) using the 'or rule'
This is P(B) × P(R) + P(R) × P(B) using the 'and rule' twice

$$\frac{8}{11} \times \frac{3}{11} + \frac{3}{11} \times \frac{8}{11}$$

$$= \frac{24}{121} + \frac{24}{121}$$

$$\frac{48}{121}$$